COLLEGE OF ENGINEERING, GUINDY

DEPARTMENT OF INFORMATION SCIENCE AND TECHNOLOGY

MINI PROJECT REPORT

SUBJECT: DATA STRUCTURES USING C

TOPIC: CANTEEN ORDER MANAGEMENT SYSTEM

SUBMITTED BY:

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1. INTRODUCTION

The Canteen Order Management System is a console-based mini project developed using the C programming language. It demonstrates how data structures can be used to organise food orders in a college canteen.

The program displays a menu containing food items and their prices. It accepts the student's name, selected food items, and quantities. The total bill is calculated, and a token number is assigned to each successfully placed order. Pending orders are stored in a circular queue and processed in the order they are received.

This project demonstrates the practical application of structures, arrays, circular queues, searching, string handling, functions, and control statements in C programming.

2. OBJECTIVES

- To display the available food items and their prices.
- To accept student details and selected food items.
- To calculate the total bill based on food prices and quantities.
- To generate a token number for each order.
- To store and display pending orders.
- To search for pending orders using token numbers.
- To process orders according to the FIFO principle.
- To understand the practical implementation of data structures using C.

3. SOFTWARE REQUIREMENTS

Programming Language: C(DATA STRUCTURE)

Compiler: GCC

Development Environment: Visual Studio Code

Execution: Command-line terminal

4. DATA STRUCTURES AND CONCEPTS USED

4.1 Structures (struct)

Structures are used to group related data under a single name. The program uses two structures: Food and Order.

The Food structure stores the food ID, food name, and price. The Order structure stores the token number, student name, selected food items, total quantity, and total bill.

4.2 Arrays

Arrays store the menu items and order details. The menu is maintained as an array of Food structures, while pending orders are stored in an array of Order structures.

The program allows a maximum of five pending orders at a time.

4.3 Circular Queue

A circular queue is used to manage pending orders. It follows the FIFO (First In, First Out) principle, meaning that the first order placed is the first order processed.

The variables front, rear, and count maintain the queue positions and the number of pending orders. The modulo operator (%) allows the array positions to be reused after orders are processed.

4.4 Searching

Searching is used to locate food items by their IDs and pending orders by their token numbers.

The searchFood() function searches for a food ID in the menu. The searchOrder() function searches the pending queue for the requested token number.

4.5 String Handling

String handling is used to manage student names, food names, and selected food item details.

The program uses string functions such as strlen(), strcat(), and strcpy() to measure, combine, and copy strings.

4.6 Functions

Functions divide the program into smaller operations, making the code easier to understand and maintain.

The program includes functions for displaying the menu, searching for food, placing orders, displaying pending orders, searching for orders, and processing orders.

4.7 Control Statements

Control statements determine the flow of program execution.

The program uses if statements for decision-making, a switch statement for menu selection, for loops to traverse arrays, while loops to accept food selections, and a do-while loop to repeat the main menu.

5. SYSTEM MODULES

5.1 Menu Display

The displayMenu() function displays all available food items with their IDs and prices. This helps users select food items before placing an order.

5.2 Order Placement

The placeOrder() function accepts the student's name and displays the menu. The user enters food IDs and quantities to select multiple items. Entering 0 finishes food selection.

The program checks whether the queue is full and validates the entered food IDs and quantities. If no food is selected, the order is cancelled.

5.3 Bill Calculation

The program calculates the cost of each selected item by multiplying its price by the quantity.

The individual item totals are added to calculate the final bill. The total quantity of all selected items is also recorded.

Formula:

Item Total = Food Price × Quantity

Total Bill = Sum of the Item Totals

5.4 Token Generation

A token number is assigned to each successfully placed order. The token counter starts at 1 whenever the program starts and increases for every new order during that execution.

The token number helps identify and search for pending orders.

5.5 Pending Order Display

The displayOrders() function displays all pending orders in the circular queue.

The displayed details include the token number, student name, selected food items, total quantity, and total bill.

5.6 Order Searching

The searchOrder() function accepts a token number and searches for the corresponding order among the pending orders.

If the token is found, the program displays the order details. Otherwise, it displays a message indicating that the pending order was not found.

5.7 Queue Management and Order Processing

The processOrder() function processes the order at the front of the circular queue.

After displaying the order details and its completed status, the function advances the front position and decreases the number of pending orders. This ensures that orders are processed in FIFO order.

6. ALGORITHM

Step 1: Start the program.

Step 2: Initialise the menu, circular queue variables, and token counter.

Step 3: Display the main menu containing the available operations.

Step 4: Accept the user's choice.

Step 5: If the choice is 1, accept the student's name, display the food menu, accept food IDs and quantities, calculate the total bill, generate a token, and insert the order into the circular queue.

Step 6: If the choice is 2, display all pending orders.

Step 7: If the choice is 3, accept a token number and search for the corresponding pending order.

Step 8: If the choice is 4, process the oldest pending order using the FIFO principle.

Step 9: If the choice is 5, display the exit message and terminate the program.

Step 10: If an invalid choice is entered, display an appropriate message.

Step 11: Repeat the main menu until the user chooses to exit.

Step 12: Stop the program.

7. SAMPLE MENU

The program contains the following food items:

Food ID	Food Item	Price
1	Dosa	Rs. 40.00
2	Idli	Rs. 20.00
3	Pongal	Rs. 35.00
4	Vadai	Rs. 10.00
5	Sandwich	Rs. 50.00
6	Samosa	Rs. 15.00
7	Juice	Rs. 30.00
8	Meals	Rs. 80.00
9	Poori	Rs. 35.00
10	Fried Rice	Rs. 70.00

8. SAMPLE OUTPUT

The following is an illustrative description of the program output. The actual values depend on the user's input.

Main Menu

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1. Place Order and Calculate Bill

2. Display Pending Orders

3. Search Order by Token

4. Process Next Order

5. Exit

Enter your choice:

Order Placement

When the user selects option 1, the program asks for the student's name and displays the food menu. It then accepts food IDs and quantities until the user enters 0.

The program displays the selected food items, total quantity, total bill, token number, and pending status. A success message confirms that the order has been placed.

Pending Orders

When option 2 is selected, the program displays the details of all orders currently waiting in the queue.

Order Searching

When option 3 is selected, the program accepts a token number and displays the corresponding pending order if it exists.

Order Processing

When option 4 is selected, the program processes the oldest pending order and displays its details with the status marked as completed.

Exit

When option 5 is selected, the program displays a thank-you message and terminates.

9. ADVANTAGES

- Simplifies food-order management in a small canteen.
- Reduces manual effort in calculating bills.
- Uses token numbers to identify orders.
- Processes orders in FIFO order.
- Supports searching for pending orders by token number.
- Demonstrates structures, arrays, circular queues, and other C programming concepts.
- Provides practical experience in implementing data structures.

10. LIMITATIONS

- The program supports a maximum of five pending orders at a time.
- Order details are stored in memory and are not saved permanently to a file.
- Token numbering starts from 1 each time the program is restarted.
- Only pending orders can be searched using the order-search function.
- Processed orders are removed from the queue and are not maintained as order history.
- The program operates through a command-line interface.

11. FUTURE ENHANCEMENTS

- Increase the maximum number of pending orders.
- Store order information in files to retain data after the program closes.
- Maintain a history of completed orders.

- Add permanent daily token management if required.
- Improve input validation and error handling.

12. CONCLUSION

The Canteen Order Management System demonstrates how fundamental data structures can be applied to a practical problem in a college canteen. Structures store food and order information, arrays hold menu and order details, and a circular queue manages pending orders using the FIFO principle.

The program also implements searching, string handling, functions, and control statements to support menu display, order placement, bill calculation, token generation, pending order display, and order processing.

Overall, this project provides practical experience in applying Data Structures concepts through a C program and improves understanding of how data structures help organise and process information efficiently.

13. REFERENCES

1. Class notes and learning materials on Data Structures Using C.
2. C programming concepts related to structures, arrays, functions, and string handling.
3. Learning materials on circular queues and FIFO order processing.